

Build with AI: Code for Communities

CivicPulse-BRICS

AI-Powered Infrastructure Governance & Citizen Triage

By Team Syntax Error



The Problem

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Fragmented feedback & blind spots in municipal infrastructure

Fragmented Citizen Reporting

Citizens have no unified, multilingual platform to report broken infrastructure (potholes, water lines). Reports get lost in bureaucracy.

Manual Triage Bottlenecks

City planners receive unstructured text and unverified photos, lacking the resources to accurately prioritize critical hazards.

Misaligned Capital Spending

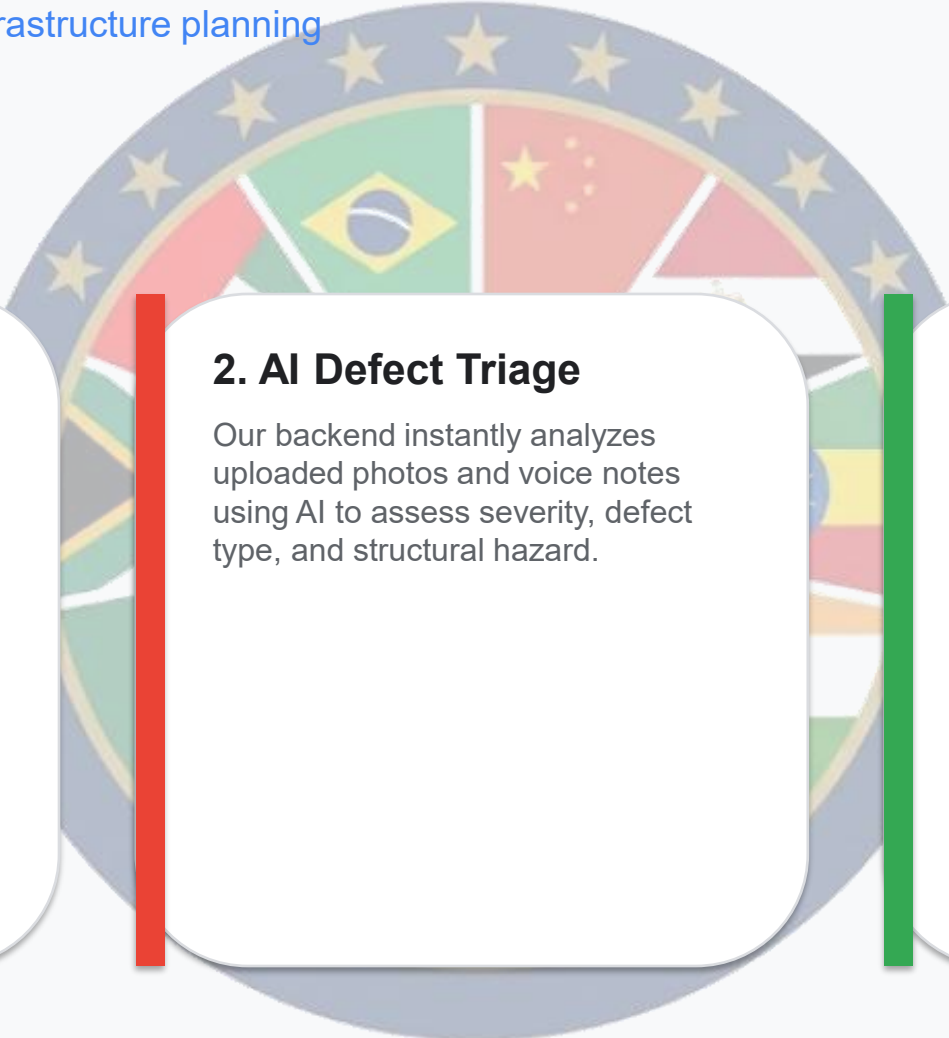
Budgets are allocated without correlating real-time citizen demand, leading to neglected hotspots and inefficient resource use.

Language & Accessibility Barriers

Diverse populations in BRICS nations are excluded from participatory budgeting due to language barriers and complex apps.

The Solution: CivicPulse-BRICS

A sovereign Digital Public Good for infrastructure planning



1. Multimodal Citizen App

Citizens can report issues using photos, voice notes, and GPS. Community demands can be upvoted for participatory budgeting.

2. AI Defect Triage

Our backend instantly analyzes uploaded photos and voice notes using AI to assess severity, defect type, and structural hazard.

3. Official Command Center

Provides city officials with an auto-prioritized repair queue, GIS heatmaps, and a CapEx budget misalignment matrix.

Our AI Approach (Google Gemini)

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How we process unstructured citizen petitions into actionable data

Gemini Vision: Infrastructure Defect Analysis

When a citizen uploads a photo of broken infrastructure, we prompt Google Gemini to analyze the image. It returns a structured JSON assessing the defect category, severity score (1-10), and potential public risk.

Gemini Text: Multilingual Translation & Policy Generation

We use AI to transcribe and translate local dialects into standard English for officials. Gemini is also used to generate strategic 'Mega Plans' by analyzing aggregated complaint data against municipal budgets.

Who It Serves

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Designed for the entire civic ecosystem

Citizens

Empowers locals to report public infrastructure issues seamlessly in their native language and upvote community demands.

Municipal Crews

Provides field teams with a prioritized, severity-ranked repair queue and direct Google Maps routing to incidents.

National Policymakers

Grants macro-level visibility into regional CapEx budget misalignment and generates high-level infrastructure investment directives.

App Feature: Citizen Gateway

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What we built for the public

Implemented Citizen Portal Features:

- GPS Auto-Detection: Instantly captures the precise location of the reported issue.
- Multimodal Uploads: Citizens can upload photos of the defect and provide descriptions.
- Community Demands Wall: A dedicated tab where citizens can propose and upvote macro-projects (e.g., new flyovers).
- Live Tracking: A tracking hub to see if their submitted complaint is 'Under Review' or 'En Route'.
- Themed UI: Fully responsive interface that works flawlessly on mobile devices.

App Feature: City Official Command Center

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What we built for government administrators

Implemented City Planner Features:

- Live Repair Queue: Automatically ingests citizen complaints and displays them with AI-assigned severity ratings.
- CapEx Misalignment Matrix: A data table cross-referencing citizen demand volumes against planned municipal budgets.
- Live Weather Telemetry: Integrated Open-Meteo API to display real-time weather conditions for the jurisdiction.
- GIS Map Integration: Leaflet.js map integration for plotting incident coordinates.
- Node Federation: Ability to toggle perspectives between different BRICS nations (India, Brazil, SA).

Demo: Citizen Gateway

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Screenshot of the Multimodal Citizen App



SCREENSHOT DEMO: CITIZEN PORTAL (INSERT HERE)

URL: <http://localhost:8000/citizen>

Demo: City Official Command Center

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Screenshot of the AI Repair Queue & GIS Tools



SCREENSHOT DEMO: CITY OFFICIAL PORTAL (INSERT HERE)

URL: <http://localhost:8000/city-official>

Demo: Sovereign Node Federation

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Screenshot of Multi-Nation CapEx Overview



SCREENSHOT DEMO: SOVEREIGN NODE HUB (INSERT HERE)

URL: <http://localhost:8000/central-official>

Deployability & How It Scales Across India

Ready for real-world municipal adoption

Technical Foundation

- Zero-Setup Backend: Built on standard Python with a lightweight SQLite database, requiring no complex infrastructure to run.
- Container Ready: Includes a Dockerfile and cloudbuild.yaml for 1-click deployment to Google Cloud Run.
- API-First: Designed with standard webhook endpoints to easily integrate with WhatsApp/Telegram bots in the future.

Scaling Across India

- Localized Deployment: Any municipality can spin up their own 'node' of the system.
- Low Latency AI: Uses Gemini Flash for instant triage of thousands of daily complaints.
- Open Digital Public Good: Adheres to DPG principles, ensuring transparent data schemas that integrate with data.gov.in.



Thank You!

Team Syntax Error

Paras Singh

Team Lead & Full-Stack

Project concept, AI system design, and core platform development.

Aradhara Pal

Frontend & UI/UX Designer

Designed website layouts, mobile screens, and dashboards.

Anshika Anand

Backend Developer

Set up the server, database integration, and Render hosting.

Himanshi

Product Research & Testing

Researched city budget data, tested user workflows, and documentation.